

ACADEMIC WORK EXPERIENCE (SELECTION)

Research Assistant

Sep 2026 - present

Department of Earth & Environmental Science, University of Pennsylvania

Philadelphia, PA

- Conduct numerical modeling and data analysis to investigate physical and biogeochemical processes in the ocean under the supervision of Prof. Irina Marinov

EDUCATION

Ph.D. in Earth and Environmental Science

2019 - 2026

University of Pennsylvania

Philadelphia, PA

Thesis: *Multi-decadal Dynamics of Southern Ocean Open-Ocean Polynyas and Deep Convection: Insights from Climate Models*

Advisor: Prof. Irina Marinov

M.Sc. in Geography

2015 - 2018

Peking University

China

Thesis: *Estimating Urban Vegetation Productivity using Chlorophyll Fluorescence Derived from Remotely Sensed Data*

Advisor: Prof. Jun Wang

B.Sc. in Environmental Science

2011 - 2015

Tongji University

China

Thesis: *Thesis: Occurrence of Heavy Metals in Marine Biota Samples in Fildes Peninsula, Antarctica*

Advisor: Prof. Qinghui Huang

PUBLICATIONS

C: Corresponding Author

PLANNED PUBLICATIONS

3. Lu, S.^C, Marinov, I. & Molotsov, S. (under review). Linking open-ocean polynyas and deep convection in the Southern Ocean across CMIP6 models. submitted to *The Cryosphere* in May 2026.
2. Lu, S., Wang, R., Marinov, I. & Molodtsov, S. (in preparation to submit to *Journal of Geophysical Research: Ocean*). Centennial Southern Ocean oscillations and circumpolar links between the Ross and Weddell Seas.
1. Lu, S., Marinov, I. & Molodtsov, S. (in preparation). Multidecadal ocean variability driven by Weddell and Ross sea convection in three generations of GFDL climate models.

PEER-REVIEWED JOURNALS

5. Wang, J., Lu, S.^C, Wang, W., Tang, L., Ma, S. & Wang, Y. (2019). Estimating vegetation productivity of urban regions using sun-induced chlorophyll fluorescence data derived from the OCO-2 satellite. *Physics and Chemistry of the Earth, Parts A/B/C*, 114, 102783. doi:10.1016/j.pce.2019.05.003
4. Du, Y., Wang, J., Lu, S., Li, J. & Cai, A. (2019). Study on Characteristics of Virtual Water Flow Spatial Change and Influencing Factors in China. *Acta Scientiarum Naturalium Universitatis Pekinensis*, 55(6), 1141. doi:10.13209/j.0479-8023.2019.117
3. Lu, S., Wang, J., Wang, Y. & Yan, J. (2018). Analysis on the variations of atmospheric CO₂ concentrations along the urban-rural gradients of Chinese cities based on the OCO-2 XCO₂ data. *Int. J. Remote Sens.*, 39(12), 4194-4213. 10.1080/01431161.2017.1415482
2. Wang, J., Wang, Y., Lu, S. & Wang, S. (2018). Large-scale assessments and mapping of snow hazards in pastoral areas of the Tibetan Plateau using microwave remote sensing data and ground records. *Nat. Hazard.*, 90(1), 461-476. 10.1007/s11069-017-3055-4
1. Yan, J., Wang, J., Lu, S. & Zeng, H. (2017). Impacts of Rapid Urbanization on Carbon Dynamics of Urban Ecosystems in Shenzhen. *Ecology and Environmental Sciences*, 26(3), 553-560.

COMMITTEE WORK

Membership

AGU, AAAS, SOCCOM

Reviewer

IPCC AR7

CONFERENCES PRESENTATION

P: poster, O: oral presentation

- 2026 ^P **Lu, S.**, Miller, L., Wang, R., Marinov, I. & Molotsov, S. Heat and carbon exchange signatures of Southern Ocean open-ocean polynyas across CMIP6 Models. *Ocean Sciences Meeting 2026*. February 22-27, 2026
- 2024 ^P **Lu, S.**, Molotsov, S. & Marinov, I. Linking multi-decadal variability of deep convection in the Southern Ocean with subpolar gyres dynamics. *SOCCOM annual meeting 2024*. June 25-27, 2024
- ^O **Lu, S.**, Molotsov, S. & Marinov, I. Local dynamics of deep convection and sea ice in Southern Ocean subpolar gyres: insights into multi-decadal open-ocean polynya variability. *Ocean Sciences Meeting 2024*. February 18-23, 2024
- ^P Molotsov, S., Marinov, I. & **Lu, S.** Teleconnections between the tropical atmosphere and Southern Ocean on multidecadal time scale. *Ocean Sciences Meeting 2024*. February 18-23, 2024
- 2023 ^O **Lu, S.**, Molotsov, S. & Marinov, I. Understand Southern Ocean Open-ocean Convection in CMIP6: GFDL-CM4 as an example. *SOCCOM annual meeting 2023*. June 7-9, 2023
- 2022 ^O **Lu, S.**, Molotsov, S. & Marinov, I. Defining open-sea convection indices for the Southern Ocean across models. *SOCCOM annual meeting 2022*. June 15-17, 2022
- ^O **Lu, S.**, Molotsov, S. & Marinov, I. Deep water formation in the North Atlantic and Southern Ocean: interhemispheric links on multidecadal timescales. *Ocean Sciences Meeting 2022*. February 24 - March 4, 2022
- 2020 ^O Marinov, I., **Lu, S.**, Sharma, P., Cabré, A. & Asadieh, B. Seasonal cycles of plankton ecology and CO₂: subpolar lessons from satellite data and coupled climate models. *Ocean Sciences Meeting 2020*. February 16-21, 2020
- ^P Kochetkova, E., Kostadinov, T., Marinov, I., **Lu, S.** & Roy, S. Seasonality of Phytoplankton in two satellite products and novel CMIP6 climate models. *AGU Fall Meeting 2020*. December 7-11, 2020
- ^P Marinov, I., Asadieh, B., Molotsov, S. & **Lu, S.** Linking High Latitude Convection with the Meridional Overturning Circulation in a Hierarchy of GFDL climate Models. *AGU Fall Meeting 2020*. December 7-11, 2020

TEACHING EXPERIENCE

COURSE TEACHING

Ocean-Atmosphere Dynamics

Fall 2024/2026

Mixed-level junior, senior, and graduate course

UPenn

- Assist with homework design and demonstrations of laboratory water-tank experiments

Climate and Big Data

Spring 2024/2025/2026

Mixed-level undergraduate and graduate course

UPenn

- Teach Python programming and data-processing workflows on HPC systems
- Mentor students on course-based research projects

Earth System Science

Fall 2023/2025

Introductory undergraduate course

UPenn

- Lead recitation sections and facilitate student discussions

Introduction to Oceanography

Spring 2021/2023

Introductory undergraduate course

UPenn

- Facilitate in-class discussions and student participation

RESEARCH MENTORING

- Mentored 7 undergraduate and 2 master's students in summer research projects, providing guidance on climate model data analysis

ADDITIONAL TRAINING (SELECTION)

CERTIFICATE

CETLI Teaching Certificate, University of Pennsylvania

2024

WORKSHOPS & SUMMER SCHOOLS

The Copernicus Arctic Hub training: data and tools for the Arctic, Copernicus Hub (online)	Sep 2026
International Workshop for OneArgo, Tohoku University (online)	May 2026
30th Annual CESM Workshop, NCAR (online)	June 2025
Trustworthy Artificial Intelligence for Environmental Science Summer School, NCAR (online)	June 2022
26th Annual CESM Workshop, NCAR (online)	June 2021
B-SOSE Workshop, SOCCOM (online)	June 2020

RESEARCH VISIT

Visiting PhD Student, School of Oceanography, University of Washington

Jan-June 2022

GRADUATE COURSEWORK

Princeton University, Atmospheric and Oceanic Sciences Programe (online class during COVID)

- Introduction to Geophysical Fluid Dynamics (Fall 2020)
- Atmospheric and Oceanic Wave Dynamics (Spring 2021)

OTHER RESEARCH EXPERIENCE

Key Laboratory for Human and Environmental Science and Technology

2015 - 2019

Guided by Prof. Jun Wang

Shenzhen, China

- Conducted a series of studies about spatial distribution of carbon emissions in Chinese cities using remote sensing data, field sampling data, and the data from Statistical Yearbook
- Practiced the use of numerical models (WRF) to simulate the influence of meteorological factors on atmosphere carbon dioxide concentration
- Processed ocean color data at China's coasts for coastal urban environment analysis

National Undergraduate Innovation Programs

2013 - 2015

Guided by Prof. Rongchang Wang from Tongji University

Shanghai, China

- Developed and tested three kinds of ion-selective microelectrode for rapid detection of Nitrate ion, Nitrite ion, and pH value in natural waters, with two kinds proved to be successfully working

MOE Key Laboratory of Yangtze River Water Environment

2012 - 2015

Guided by Dr. Penghui Li & Prof. Qinghui Huang

Shanghai, China

- Processed water samples from Yangtze Estuary and conducted analyses of dissolved organic matter (DOM) using 3-D fluorescence spectroscopy
- Tested the toxicity of pollutants on algae using ToxY-PAM, analyzed the compound of pollutants using UV-visible spectroscopy

SKILLS

Programming & Scientific Computing

Python (NumPy, SciPy, Pandas, Xarray, Dask), MATLAB, Julia;
familiar with R, FORTRAN

Visualization & Geospatial Analysis

Matplotlib, Cartopy, Seaborn, ArcGIS/QGIS

HPC & Software Development

Linux/Unix, SLURM/PBS, Git/GitHub, Conda

Scientific Communication

LaTeX, Typst, Markdown/Quarto, Jupyter Notebooks, Julia Pluto